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BME 60D

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Tutorial 9 – Beyond Ansys: putting our skills to the test in other packages

This tutorial replicates tutorial 1 of bending a simply supported steel beam under a central load, originally performed in Ansys Workbench, now repeated in Abaqus/CAE Learning Edition. The goal was to determine whether comparable results could be achieved using Abaqus.

In Ansys, the beam ($1000 \text{ mm} \times 40 \text{ mm} \times 40 \text{ mm}$) was subjected to a $2,000 \text{ N}$ downward force over a $20 \text{ mm} \times 40 \text{ mm}$ area at its center. Boundary conditions consisted of a pinned support at one end and a roller support at the other. Using Structural Steel ($E = 200,000 \text{ MPa}$, $\nu = 0.3$), the maximum stress was $\sim 24.3 \text{ MPa}$, and deflection was approximately 0.25 mm .

In Abaqus, the same setup was modeled with coarse meshing (element size = $10 \rightarrow 40 \text{ mm}$) to stay within the $1,000$ node limit. The maximum Von Mises stress obtained was $\sim 20.3 \text{ MPa}$, located at the center of the beam. The displacement magnitude was slightly higher due to mesh coarseness but remained within the same order of magnitude. The differences are primarily due to fewer elements, which equals less resolution near the load application zone and slightly different interpolation methods in Abaqus. Overall, Abaqus replicated the stress pattern and magnitude accurately, confirming theoretical consistency.

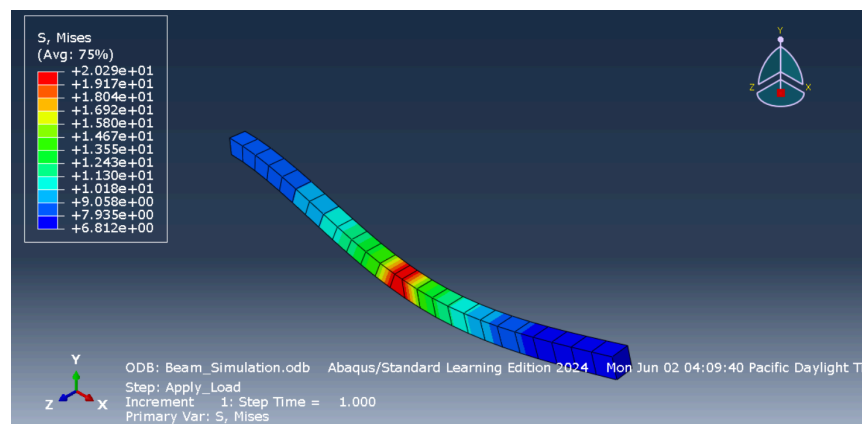


Figure 1. Von Mises stress distribution in a simply supported steel beam under a 2000 N central load, simulated in Abaqus. Peak stress ($\sim 20.3 \text{ MPa}$) occurs at the center of the beam, as expected from theoretical bending behavior.

Using Abaqus was initially more challenging than Ansys due to its less guided interface. Abaqus requires manual transitions between modules (Part, Property, Assembly, Step, Load, Mesh, Job), which adds complexity compared to Ansys Workbench's integrated workflow.

However, once the learning curve was overcome, the process became intuitive. I gained confidence in navigating Abaqus's logic after understanding how instance types (independent vs. dependent) affect meshing and how boundary conditions and loads must be reapplied after remeshing. I now feel confident that I could adapt to other non-Ansys packages with documentation and practice. If hired by a company using Abaqus or FEBio, I would be capable of performing meaningful simulations after a brief ramp-up.

Ansys Workbench and Abaqus CAE differ significantly in both workflow and advanced capabilities. Ansys Workbench offers a highly intuitive user interface with integrated geometry creation via SpaceClaim or Discovery, making it particularly beginner-friendly. Its meshing process is largely automated, with built-in refinement options that simplify setup. Boundary conditions in Ansys are applied through a drag-and-drop graphical interface, and its solvers are fast and GPU-optimized, which enhances computation speed. While scripting in Ansys is limited, it is generally sufficient for standard applications.

In contrast, Abaqus CAE requires more manual setup, with geometry created through sketching or external CAD imports, and meshing involves manual seeding and configuration. However, Abaqus excels in flexibility—it supports a wide range of material models, including nonlinear, hyperelastic, anisotropic, and even user-defined behaviors via UMAT subroutines. Its boundary conditions and loads are applied through structured modules, which can feel less streamlined but offer more control.

In terms of biomedical relevance, Abaqus excels in support for advanced constitutive models (e.g., anisotropy, viscoelasticity) and contact mechanics—important for modeling tissues, implants, or interactions in orthopedics and cardiovascular devices.

If I were starting a biomedical engineering company that uses FEM to aid in product development (e.g., stents, prosthetics, tissue scaffolds), I would choose Abaqus as the core simulation package. Although Ansys is more beginner-friendly and faster for general use, Abaqus provides superior flexibility, scripting capability, and advanced material modeling—all critical for accurate, physiology-based models.

With this project, I've demonstrated that even with a limited license and reduced mesh resolution, Abaqus can match the predictive accuracy of Ansys when properly configured. While each tool has strengths, Abaqus offers more depth for research-driven, high-fidelity biomedical simulation.